HGWM: Hierarchical Graph-Guided World Model for Zero-Shot Object Navigation via Graph Matching

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Project page Code

Abstract

Zero-shot object navigation requires an embodied agent to choose where to search while retaining evidence from previous observations. HGWM couples an LLM-generated room–anchor prior with persistent spatial-semantic memory. A coarse stage assigns visual-semantic scores to candidate headings on a 0–10 scale; a fine stage measures symbolic compatibility between observed objects, room context, and the goal prior. Phase-dependent fusion combines these scores with an LLM direction recommendation before local action selection. On HM3D v0.1, the reported HGWM configuration achieves 62.6% success rate (SR) and 31.5% success weighted by path length (SPL), compared with 58.1% and 31.2% for WMNav. On MP3D, the corresponding values are 46.8% and 17.9%, compared with 45.4% and 17.2%. The results indicate higher success point estimates and modest path-efficiency gains. On HM3D v0.2, the full model obtains 74.4% SR and 31.9% SPL. Additional component ablations reduce SR by 0.9–2.3 percentage points and SPL by 0.6–1.5 points. These are author-reported aggregate measurements. The accompanying implementation repairs pass offline tests; episode-level rerun records are not available. Evaluation is confined to simulation.

1 Introduction

Navigation in an unfamiliar environment benefits from both recognizing visible objects and anticipating where a target might be found. Foundation-model agents provide open-vocabulary semantic cues [17, 26], but ambiguous observations and incomplete spatial memory can still lead to unproductive search. A useful prior should guide exploration without being mistaken for a verified map of the environment.

We study two design challenges: retaining observations across viewpoints and using prior room–object relations to assess candidate directions. VLM detections can be incorrect, while common indoor associations need not hold in every scene. Structural checks may therefore help reject some misleading interpretations but can also suppress valid, atypical observations.

HGWM addresses these challenges through a *predict-then-ground* design. An LLM first generates a hierarchical goal graph describing plausible rooms, anchors, and connections. During navigation, RGB-D observations populate a persistent spatial-semantic graph. Coarse visual-semantic scoring and fine prior-conditioned matching then inform action selection. Here, grounding means relating a hypothesis to sensor-derived evidence, not proving that either graph is correct.

A category such as bed specifies no room instance, route, or viewpoint. A room–anchor hypothesis supplies plausible search contexts before the target appears, while memory records encountered evidence. Their compatibility gives relational context a defined role in ranking where to look next.

Our contributions are threefold:

- A symbolic predictive navigation state coupling a hierarchical goal prior with persistent spatial-semantic graph memory.
- A two-stage decision rule that separates prior-conditioned visual scoring from explicit, direction-specific matching of observed and hypothesized relations.
- Evaluation on HM3D and MP3D, including four additional component ablations on HM3D v0.2, analysis of success and path efficiency, and qualitative navigation examples.

Relative to the reported WMNav results, HGWM improves SR by 4.5 percentage points (pp) on HM3D v0.1 and 1.4 pp on MP3D; SPL improves by 0.3 pp and 0.7 pp, respectively. These are point-estimate comparisons, not evidence of statistical significance. The qualitative trajectories motivate studying repeated exploration, but no aggregate loop-frequency metric is reported.

2 Related Work

Semantic search and spatial maps. ObjectNav methods differ in how they represent prior knowledge and retain observations [13, 20]. Habitat-Web and OVRL-V2 learn navigation capabilities from task data [19, 24]; ZSON, CoW, ESC, OpenFMNav, and semantic-frontier approaches exploit pretrained representations or commonsense knowledge [6, 9, 15, 16, 35]. VLFM accumulates visual-language relevance in a frontier value map [28]. VoroNav uses a reduced Voronoi graph and path descriptions for waypoint reasoning [22], while BeliefMapNav estimates target-location belief in a 3D voxel

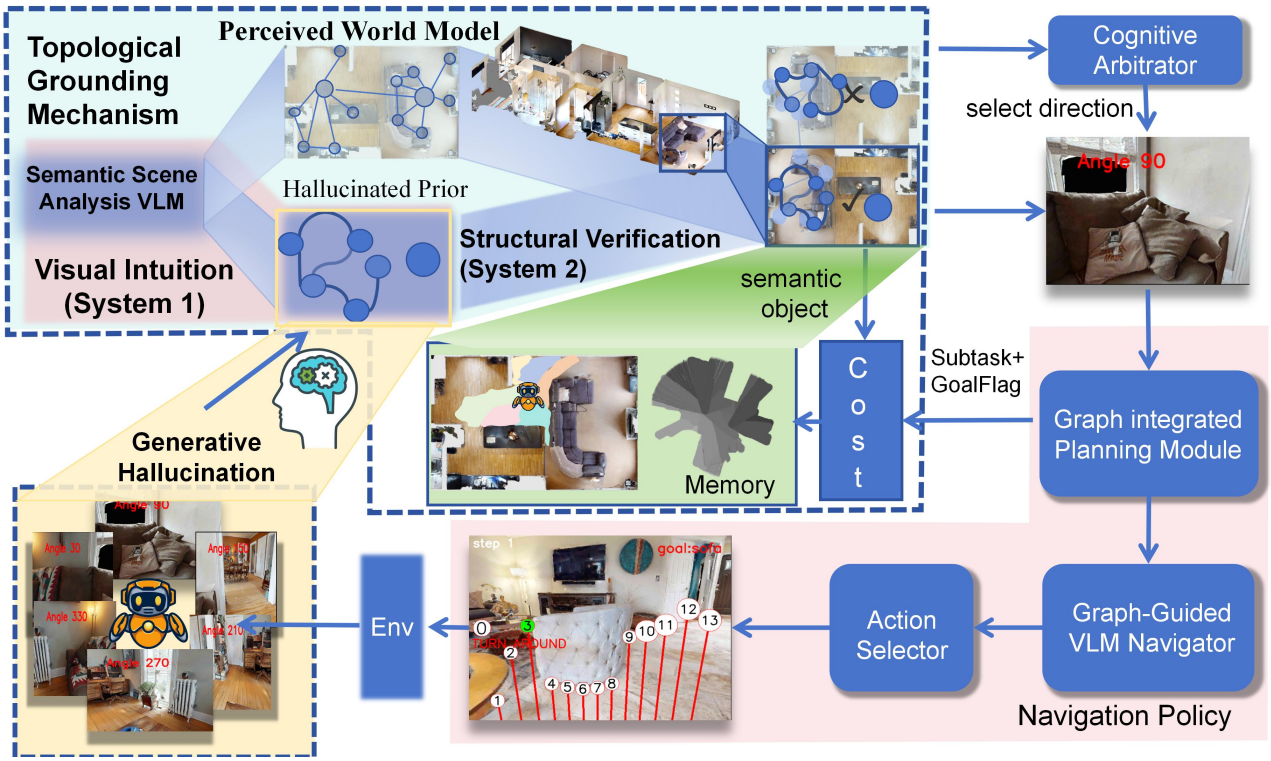


Figure 1: HGWM architecture. The LLM generates a hypothetical room–anchor prior, observations populate spatial-semantic memory, and graph-guided reasoning informs navigation. Hallucinated Prior and Structural Verification denote prior generation and heuristic compatibility checks; neither establishes scene truth. System 1/2 labels are conceptual analogies.

space and plans sequences of destinations [36]. Together with 3D-Aware and ApexNav [30, 31], these systems establish spatial memory and target-directed search as important design choices. HGWM focuses on a complementary question: how a category-conditioned relational hypothesis should influence the score assigned to each candidate heading.

Scene graphs for embodied reasoning. SOON, HOZ, and ConceptGraphs use structured scene representations [10, 32, 37]. SG-Nav builds an online hierarchy of objects, groups, and rooms, uses hierarchical language-model reasoning, and introduces graph-based re-perception [25]. HGWM therefore builds on an established use of graphs for context and perception review. Its specific design couples observed memory with a separate room–anchor hypothesis and makes their compatibility an explicit decision signal alongside visual-semantic scores. The contribution concerns this prior-to-action interface, rather than scene-graph construction alone.

Goal graphs and the ObjectNav setting. UniGoal is the closest graph-matching reference [26]. It unifies object, image, and text goals, matches node and edge embeddings, evaluates graph-edit similarity, and switches exploration strategies according to the matching state. Its paper explicitly notes that an ObjectNav category becomes a single goal node, disabling goal-graph decomposition and anchor-pair alignment in that setting. HGWM expands the category into room hypotheses, anchor objects, and relations before observation. This expansion supplies relational search cues even when the requested goal is only an object category. HGWM then scores headings with explicit symbolic features and phase-dependent fusion; UniGoal uses matched graph structure and, where applicable, anchor alignment to generate exploration destinations. Both use graph matching and staged reasoning, but they differ in the category-goal representation and the mechanism connecting matching to navigation.

VLM decision memory. WMNav scores panoramic views, accumulates a curiosity value map, and combines subtask feedback with a two-stage action proposer [17]. HGWM retains this useful map-and-policy structure while adding a separate goal prior, persistent relational evidence, and a fine compatibility channel. Its decision rule combines visual promise, graph compatibility, and an LLM direction recommendation using coefficients selected from the current matching state. This specifies how graph context changes the input to the stored direction-value map; it does not treat memory or subtask decomposition as new in themselves.

Predictive models and HGWM's contribution. SGM predicts missing regions of semantic maps [33]. World Models, Dreamer, Pathdreamer, and Navigation World Models learn predictive representations of environments [2, 11, 12, 14]. HGWM uses pretrained models to form a symbolic hypothesis

and compare it with observed evidence, without training a dynamics model or synthesizing future observations. Predictive coding is a conceptual motivation [7, 8]. The methodological contribution is the combination of category-to-graph expansion, direction-specific compatibility, and phase-dependent fusion within persistent decision memory. The experiments distinguish whole-configuration performance from evidence about each component.

3 The Proposed Method

HGWM maintains a symbolic predictive navigation state: a goal prior, an observation memory, and scores for candidate headings. The term *world model* denotes this state, not a learned dynamics simulator. Figure 1 shows the perception–memory–decision loop. Section 4.2 identifies the reported evaluation and available reproducibility records.

3.1 Task and Decision State

The target is an object category c. At decision step t, the agent receives RGB-D observations $O_t = (I_t, D_t)$ and a planar pose $P_t \in SE(2)$. The panoramic scorer uses six relative headings $\mathscr{A} = \{30, 90, 150, 210, 270, 330\}^\circ$. A high-level decision selects a search direction and a short-horizon subtask; the local navigator then selects a feasible displacement. The 40-step horizon counts agent decision iterations, not a uniform metric distance traveled.

$$\mathscr{W}_t = (G_{goal}, M_t, \{(c_t^\alpha, f_t^\alpha)\}_{\alpha \in \mathscr{A}}). \tag{1}$$

Here $c_t^\alpha \in [0, 10]$ is a coarse score and $f_t^\alpha \in [0, 1]$ is a symbolic compatibility score. Neither is a posterior probability. Success requires the stopping condition within 1.0 m of a target instance under the evaluation protocol, rather than complete alignment of the graphs.

3.2 Goal Prior and Observation Memory

Given c, an LLM returns a structured prior containing likely rooms, alternative rooms, anchor objects, and room connections:

$$G_{goal} = \mathrm{LLM}_{prior}(c, \mathrm{prompt}). \tag{2}$$

This graph is a category-conditioned hypothesis, not a map of the particular scene. In the supplied implementation, priors are cached by goal within an episode and the cache is cleared on reset. We do not assume identical priors across independently executed episodes.

What is predicted. The prior describes object categories and plausible relations, rather than asserting coordinates for unseen objects. For example, a bed query can suggest a bedroom and nearby furniture as search context. A detected anchor contributes evidence for a direction even before the bed is visible; failure to observe that anchor is not proof that a bed is absent. Alternative rooms in the prior preserve multiple search hypotheses. Hypothetical nodes are kept separate from observation records so that an expected object is not counted as a detected object merely because the LLM mentioned it.

Spatial-semantic memory stores object names, inferred rooms, observation counts, and estimated map positions. The revised implementation queues depth-based navigation rays before the VLM response arrives, then associates each directional description with the nearest available view-center ray within 30°. Projection uses pose, a maximum ray distance of 3 m, and 100 map pixels per meter. These are coarse positional estimates for a view, not precise object bounding-box projections. The current local graph additionally uses textual spatial and depth cues.

For each direction, current observations and retrieved memory are normalized into one graph contract before matching. It exposes consistent flat and typed views of room nodes, object nodes, adjacency, object relations, and containment. Field aliases are resolved once, objects retain stable identifiers, and observed room membership supplies containment edges. Current attributes take precedence for the same identifier. This boundary prevents a scorer from silently reading a different graph representation. Matching uses symbolic features, without a learned graph-embedding network.

3.3 Coarse Visual-Semantic Scoring

The VLM receives the panorama, goal, and prior context and returns a JSON record per heading with a score, objects, room, path assessment, and explanation:

$$c_t^\alpha = \mathrm{VLM}_{score}(I_t^{pano}, c, G_{goal})_\alpha. \tag{3}$$

The requested score range is 0–10. A score of 10 is reserved for a visible target according to the model; 0 denotes a dead end with no visible target; 1–9 encodes intermediate opportunities. This is a model judgment, not a validated detector confidence. Because the same goal prior already enters this prompt, the fine stage adds an explicit computation rather than exclusive access to prior knowledge. The parser supplies 5 when a score is absent, and uses score 5 for all headings if the entire directional extraction fails. The fusion routine consumes this scale directly: it does not apply candidate-set min–max normalization or episode-level z-score normalization. Malformed or out-of-range responses remain an implementation concern, not an assumed probabilistic guarantee.

3.4 Fine Symbolic Compatibility

The refinement prompt asks the LLM to review relations using scene descriptions and goal context. The application routine retains nodes, accepts proposed relations between known nodes, and preserves unmodified edges with confidence greater than 0.7. All typed edge views are rebuilt from the edited edge set, preventing discarded relations from surviving in a stale representation. This operation remains heuristic and can mishandle atypical arrangements; it is not a verified node-pruning procedure.

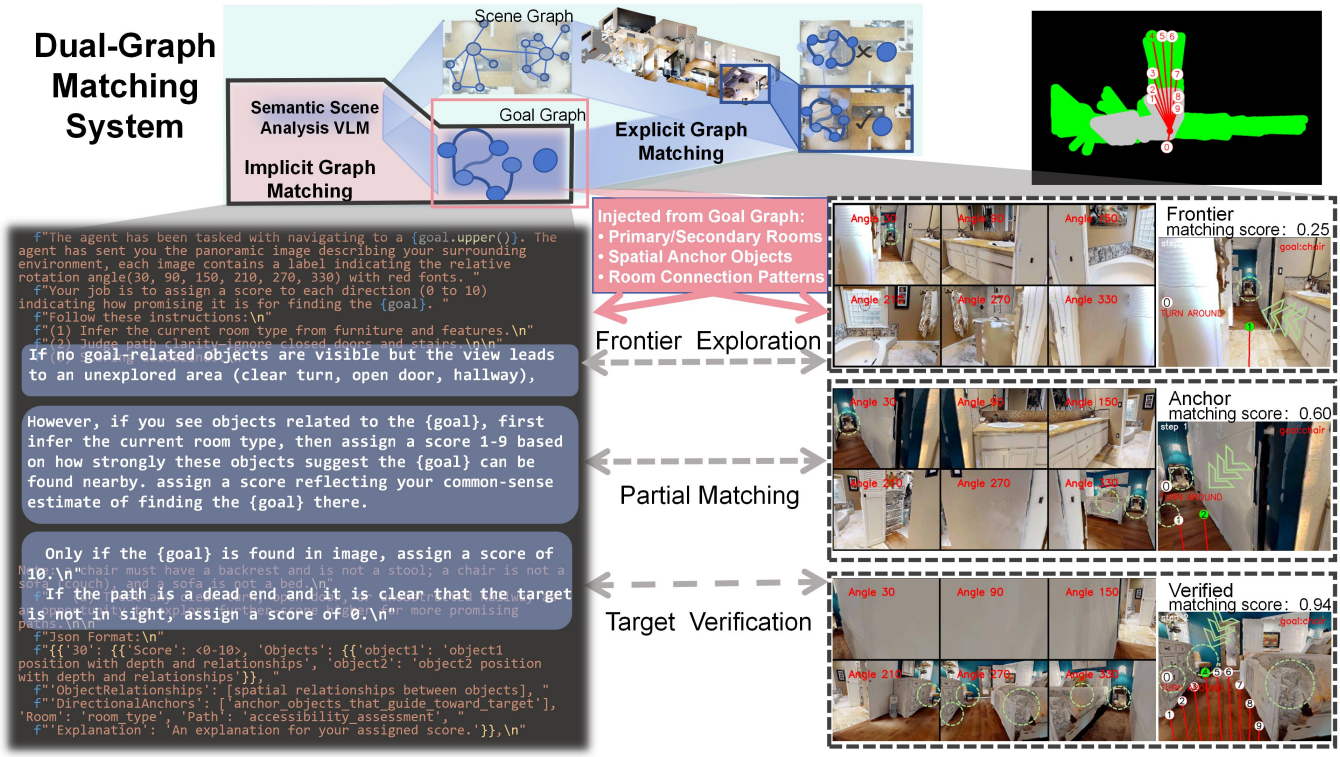


Figure 2: Coarse-to-fine illustration with example trajectory views. Coarse VLM scores use 0–10: 0 for a dead end without a visible target, 1–9 for intermediate opportunities, and 10 for a model-reported visible target. The example fine scores are heuristic compatibility values, not calibrated probabilities. The ordered phase thresholds are defined in the text.

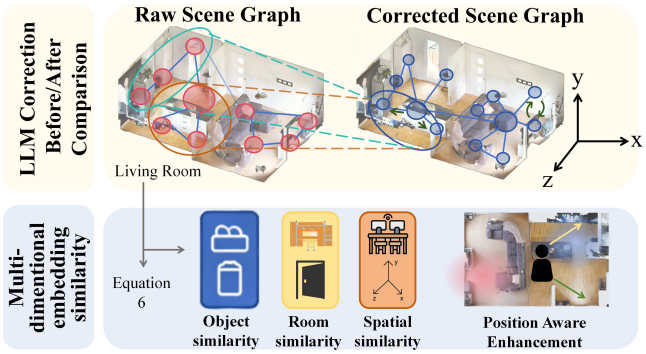


Figure 3: Fine-matching illustration. Corrected Scene Graph denotes the graph after heuristic relation proposals, not verified truth. The illustrated embedding similarity is interpreted as symbolic feature comparison, without a learned graph-embedding network. Equation 6 specifies all five features, including semantic and depth terms in addition to the illustrated object, room, and spatial terms.

Let $\mathbf{s}^\alpha = (s_r, s_o, s_p, s_m, s_d)$ denote five features: room, object, spatial, semantic, and depth. The code compares names, sets, and relations. The room feature combines Jaccard overlap of room names (0.5), overlap of spatial regions (0.3), and connectivity (0.2). The object feature compares visible objects and stored objects with the prior, and with each other. It also rewards matching anchors and rooms. Spatial compatibility combines room relations (0.4), object relations (0.4), and relations between levels (0.2). Each term compares canonical endpoint names and relation labels, dividing the intersection size by the larger relation-set size; absent evidence gives zero. Semantic consistency combines agreement of visible and stored objects (0.4), graph richness (0.3), and goal relevance (0.3). The supplement defines the bonuses, defaults, and weight cases. For name and region sets, the overlap convention is

$$J(A,B) = \frac{|A \cap B|}{\max\{1, |A \cup B|\}}. \tag{4}$$

For depth-layer proportions $\mathbf{p}$ and reference proportions $\mathbf{q} = (0.3, 0.4, 0.3)$, define

$$s_d = \frac{1}{3} \sum_{k=1}^{3} (1 - |p_k - q_k|). \tag{5}$$

Missing depth observations give $s_d = 0$. The fine score is

$$f_t^\alpha = \min\{1, (\mathbf{w}^\top \mathbf{s}^\alpha + b^\alpha) d^\alpha h^\alpha\}. \tag{6}$$

Weights depend on detected prior-associated objects, scene quality, and room/object match scores. The default vector in room, object, spatial, semantic, and depth order is $(0.35, 0.30, 0.20, 0.10, 0.05)$. The bonus b^α rewards partial matches and reliable relations. d^α is direction relevance of nearby stored observations, and h^α is a depth-direction factor;

Table 1: Fusion coefficients in the supplied implementation.

Phase	a (coarse)	b (fine)	e (LLM)
Frontier exploration	0.6	0.1	0.5
Focused search	0.6	0.4	0.4
Target verification	0.8	0.5	0.2

both default to 1 when the corresponding memory is unavailable. There is no additional multiplicative PositionScore in this inspected fusion path. An exception in overlap calculation returns $f_t^\alpha = 0.5$ and records its cause. Empty observations receive no positive graph-match evidence. Per-direction diagnostics record graph sizes, feature scores, and fallback status; the score remains a heuristic rather than a calibrated probability.

3.5 Phase-Dependent Fusion

Let $\bar{f}_t$ and f_t^{max} be the mean and maximum fine scores across the six headings. Target verification takes precedence when $f_t^{max} \geq 0.7$; otherwise focused search applies when $\bar{f}_t \geq 0.4$ or $f_t^{max} \geq 0.5$; frontier exploration applies otherwise. The LLM receives the directional scene descriptions, scores, and phase, and returns a recommended direction $\hat{\alpha}_t$ and a scalar u_t. The scalar is a reasoning output, not a calibrated confidence interval. Its contribution is

$$\ell_t^\alpha = \begin{cases} u_t, & \alpha = \hat{\alpha}_t, \\ \max(0.2, 0.5u_t), & \text{otherwise.} \end{cases} \tag{7}$$

For $c_t^\alpha < 10$, the fusion rule is

$$q_t^\alpha = \text{round}_2\Big(\text{clip}_{[0,10]}(ac_t^\alpha + 10bf_t^\alpha + 10e\ell_t^\alpha)\Big). \tag{8}$$

If $c_t^\alpha \geq 10$, the implementation sets $q_t^\alpha = 10$. Table 1 gives the actual fusion coefficients. They are heuristic coefficients, not convex-mixture probabilities; clipping is therefore part of the definition.

Prior information versus explicit matching. Prompting can exploit relations implicitly. HGWM additionally compares named endpoints and relation labels, so equal object sets can receive different spatial scores when their relations differ. At a fixed state with $c_t^\alpha < 10$, hold the prior, graph, coarse scores, phase coefficients, and LLM output constant. Replacing the fine contribution by an equal-coefficient coarse contribution gives the unclipped diagnostic

$$S_{\text{ref}}^\alpha = (a+b)c_t^\alpha + 10e\ell_t^\alpha, \qquad S^\alpha - S_{\text{ref}}^\alpha = b(10f_t^\alpha - c_t^\alpha).$$

This separates the numerical matching channel from prior access and coefficient mass. It is an algebraic diagnostic, not a measured navigation ablation; phase selection and LLM outputs can themselves depend on matching. The supplement defines controls for these distinct effects.

3.6 Memory Update and Local Action Selection

Fused directional values update the spatial curiosity map. Regions shared by adjacent headings receive the mean of their values; the base policy merges them with stored map values and selects the heading with the greatest regional mean. Thus selecting the largest instantaneous q_t^α alone does not fully specify the policy. A planning VLM uses the selected view, previous subtask, goal, and graph context to propose the next subtask and goal flag. The local action selector reasons over annotated feasible displacements; depth-derived navigability constrains motion. Stopping and navigation remain model-driven decisions and can fail despite a high score. The supplementary source map identifies the exact routines, fallback paths, and known interface issues.

More precisely, let $V_t(x)$ denote the stored curiosity value at map cell x, and let $v_t(x)$ be the fused value projected from the current panorama. For cells updated by the base policy, the merge is

$$V_t(x) = \min\{V_{t-1}(x), v_t(x)\}. \tag{9}$$

Cells outside the update mask retain their previous values; cells in the goal mask are assigned 10. A valid directional region is ranked by its mean stored value. This inherited conservative update makes the decision depend on previously accumulated evidence, not just the latest panorama. It can also preserve an erroneous low score, which motivates evaluating failures at the trajectory level.

3.7 Execution and Information Flow

Each navigation decision follows the same information flow:

1. Obtain the panorama, depth, and pose; generate or reuse the goal prior for the current episode.
2. Request the six visual direction scores together with object, room, and path descriptions.
3. Associate semantic descriptions with spatial memory and form direction-specific graph evidence. Where enabled, request relation proposals before computing compatibility features.
4. Compute the fine scores, choose the exploration phase, and obtain the joint LLM direction recommendation. Apply the phase coefficients and score clipping.
5. Update the curiosity map, select a direction, and use the chosen view and previous subtask to produce the next local objective.
6. Select a feasible displacement from the annotated action candidates, execute it, and use the next observation to continue or stop.

This ordering separates the goal hypothesis, observed evidence, direction ranking, and motion interface. It also identifies what an ablation changes: the prior affects contextual conditioning; relation refinement affects graph evidence; arbitration affects the fused score; and weight adaptation changes the fine feature

mixture. None of these changes requires treating an LLM score as a calibrated probability.

4 Experiments

We report simulation experiments on HM3D and MP3D to evaluate the complete HGWM configuration. The experiments do not include physical-robot trials or a systematic sensor-noise robustness protocol.

4.1 Datasets and Evaluation Metrics

Datasets. We evaluate on HM3D v0.1, HM3D v0.2, and MP3D [5, 18]. We report main comparisons on HM3D v0.1 for consistency with prior work, and use HM3D v0.2 for the three-configuration component comparison. Results from these splits should not be pooled as if they used the same episodes.

Evaluation metrics. We report Success Rate (SR) and Success weighted by Path Length (SPL), following standard ObjectNav evaluation protocols [1, 3]. These metrics measure navigation effectiveness and efficiency, respectively. Figure 4 illustrates selected trajectories; repeated exploration is not reported as a quantitative metric.

Reading SR and SPL together. For N episodes, let s_i indicate success, l_i be the shortest valid path length, and p_i be the traveled path length. The reported percentage metrics are

$$\begin{aligned} \mathrm{SR} &= \frac{100}{N}\sum_{i=1}^{N} s_i, \\ \mathrm{SPL} &= \frac{100}{N}\sum_{i=1}^{N} s_i \frac{l_i}{\max(l_i, p_i)}. \end{aligned} \tag{10}$$

SR measures completion frequency; SPL also penalizes excess travel on successful episodes. A higher SR therefore need not imply a higher SPL. All differences below are absolute percentage-point changes, computed within a dataset and split; HM3D v0.1 and v0.2 results are kept separate.

4.2 Implementation and Reproducibility

The agent is configured for at most 40 navigation steps per episode, with a cylindrical body of radius 0.18 m and height 0.88 m. Its RGB-D camera has a resolution of 640×480, a 79° horizontal field of view, and a 14° downward pitch. The success threshold is $d_{thres} = 1.0$ m. The supplied configuration uses a navigation-map scale of 100 pixels per meter.

Gemini-1.5-Pro supplies scene understanding and coarse scores, while Qwen2.5-7B-Instruct supplies goal priors, relation proposals, and direction recommendations. The accompanying source documents prompts and episode-local prior caching. We could not establish persistent identical-query VLM caching or fixed cross-episode priors from that snapshot. The main comparisons retain the original reported measurements, and the additional HM3D v0.2 component results are specified in Table 5. We report aggregate results and a three-run range for the HM3D v0.1 main setting; episode-level outcomes, paired uncertainty, and per-variant query totals are unavailable. The author reports unchanged performance after the repairs, with corrected voxel-memory and no-refinement SPL entries. Source-linked episode records are not available for independent verification; local validation covers offline software tests. New runs record source/configuration hashes, episode identities, client-call counts and timings, and explicit episode outcomes to support a matched rerun.

4.3 Main Results

Table 2 reports HGWM at 62.6% SR and 31.5% SPL on HM3D v0.1, compared with WMNav's 58.1% and 31.2%: gains of 4.5 pp SR and 0.3 pp SPL. On MP3D, HGWM reports 46.8% SR and 17.9% SPL versus 45.4% and 17.2%, giving 1.4 pp SR and 0.7 pp SPL. In particular, the path-efficiency gains are modest. OVRL-V2 has a higher reported HM3D SR (64.7%) but uses a different training regime.

Across three HGWM runs on HM3D v0.1, SR ranges from 62.3 to 62.8 and SPL from 31.2 to 31.7. The tabulated values are reference points; their status as run means is not specified. These observed ranges are not confidence intervals. The SPL minimum equals the reported WMNav point estimate, so a consistent SPL advantage is not established. Paired baseline runs and equal-query comparisons are not supplied.

Table 3 makes the magnitude and direction of the reported differences explicit without treating them as additional experiments.

4.4 Qualitative Navigation Behavior

Figure 4 shows selected WMNav and HGWM trajectories. The supplement additionally contains all 18 supplied historical case folders, indexing 338 step directories and 337 text records. A common first/middle/last rule selects original views and maps for an offline gallery; all available step logs are retained. These cases expose prompts, responses, and decisions for inspection. They are neither a paired comparison nor a rerun of the repaired implementation, and model goal flags are not treated as simulator success labels. Aggregate loop-frequency and robustness claims therefore remain outside this qualitative evidence.

4.5 Backbone Sensitivity

Table 4 compares two VLM settings on HM3D v0.1. Gemini-1.5-Pro reports 62.6% SR and 31.5% SPL; Gemini 2.0 Flash reports 59.2% and 30.8%, drops of 3.4 pp and 0.7 pp. This is a limited sensitivity check, not a systematic test across model families or sensor conditions. No component-matched baseline or perception-error measurements are available to establish why performance changes or whether graph matching compensates for backbone errors.

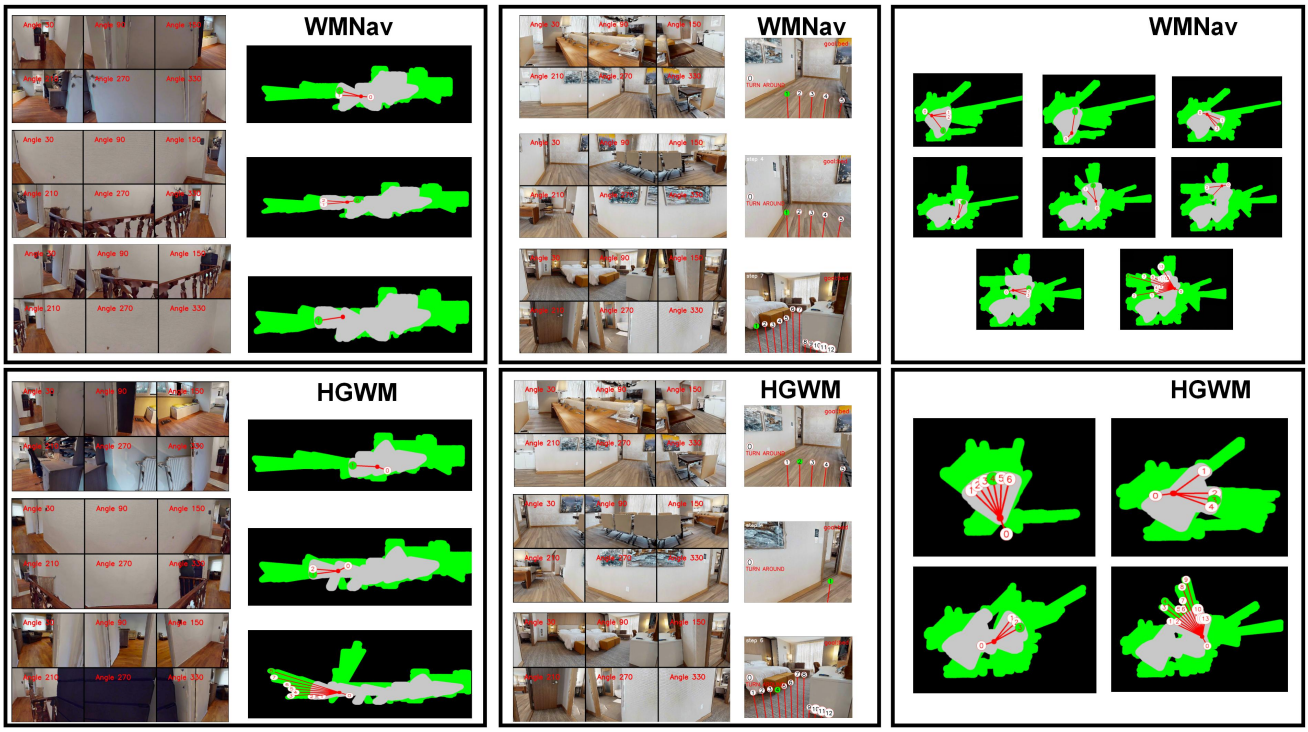


Figure 4: Decision process comparison between WMNav and HGWM. In these selected examples, HGWM visits different regions and exhibits less repeated exploration. These trajectories illustrate behavior; they do not establish an aggregate reduction in looping or isolate the effect of graph matching.

4.6 Component Analysis on HM3D v0.2

Table 5 combines the original memory configurations with four additional component results. The complete HGWM configuration is the common reference at 74.4% SR and 31.9% SPL. Basic VLM navigation obtains 67.8/29.3, while adding voxel memory gives 70.9/30.1. The latter remains the original voxel-memory baseline; a separate graph-memory configuration with direct VLM scoring has not been identified by these measurements.

Prior and relation evidence. Removing the goal prior yields 72.1% SR and 30.8% SPL, decreases of 2.3 and 1.1 points. Disabling relation refinement gives 72.1% SR and 30.4% SPL, decreases of 2.3 and 1.5 points. The larger SPL drop supports the contribution of relation processing at the aggregate level. The prior supplies scoring context; refinement changes matching evidence. Paired trajectories are needed to determine which episodes benefit.

Meaning of the interventions. The no-prior control retains the category but substitutes an unknown room, empty connections, and a minimal goal anchor; pretrained commonsense remains available. No refinement bypasses relation edits. The historical no-arbitration control removes the LLM contribution without renormalizing the remaining coefficients, changing both information and score scale. The weight control affects the five fine features, not the phase-dependent fusion coefficients. Its historical exact mode is unspecified; the revised runner names static and uniform modes separately and records their vectors.

Arbitration and adaptive weighting. Without LLM arbitration, performance is 72.7/31.1, below the full model by 1.7 SR and 0.8 SPL points. The non-adaptive fine-weight control gives 73.5/31.3, decreases of 0.9 and 0.6 points. The point estimates therefore favor the full fusion design over these reported alternatives. Their relative drops should not be interpreted as additive contributions: each intervention is evaluated against the full configuration, and joint removals are not reported. This historical result does not establish separate performance for the two weight choices.

Controls prepared for reevaluation. The revised code adds a scale-matched no-arbitration control with

$$(a',b',e') = \frac{a+b+e}{a+b}(a,b,0).$$

It preserves the original coefficient mass while removing the LLM term. Additional named controls retain the same prior prompt with no graph scoring or planning context, or retain graph memory with direct directional scoring. Their outcomes are not separately reported; the existing no-arbitration result uses unnormalized removal. Neither control isolates matching alone: the implemented flags also change graph context,

Table 2: Reported comparison on HM3D v0.1 and MP3D. SR and SPL are percentages; higher is better. Backbone, training, and evaluation settings differ across methods; this is not a controlled component comparison. For HGWM on HM3D v0.1, superscripts/subscripts denote the observed maximum/minimum deviations across three runs, not confidence intervals. Other entries are point estimates; methods were not rerun in a shared implementation.

Method	Vision	Language	HM3D v0.1		MP3D	
			SR↑	SPL↑	SR↑	SPL↑
Habitat-Web [19]	-	-	41.5	16.0	31.6	8.5
OVRL [23]	-	-	-	-	28.6	7.4
OVRL-V2 [24]	-	-	64.7	28.1	-	-
ZSON [16]	CLIP	-	25.5	12.6	15.3	4.8
PSL [21]	CLIP	-	42.4	19.2	18.9	6.4
PixNav [4]	LLaMA-Adapter	GPT-4	37.9	20.5	-	-
SGM [33]	-	-	60.2	30.8	37.7	14.7
VLFM [28]	BLIP-2	-	52.5	30.4	36.4	17.5
CoW [9]	CLIP	-	-	-	9.2	4.9
ESC [35]	-	GPT-3.5	39.2	22.3	28.7	14.2
L3MVN [29]	-	GPT-2	50.4	23.1	-	-
VoroNav [22]	BLIP	GPT-3.5	42.0	26.0	-	-
HyPerNav [27]	Qwen-VL-7B-Instruct	-	56.3	29.7	43.8	16.9
TopV-Nav [34]	-	-	45.9	28.0	31.9	16.1
SG-Nav [25]	LLaVA-1.6-7B	GPT-4	54.0	24.9	40.2	16.0
UniGoal [26]	LLaVA-1.6-7B	LLaMA-2-7B	54.5	25.1	41.0	16.4
WMNav [17]	Gemini-1.5-Pro	-	58.1	31.2	45.4	17.2
HGWM (Ours)	**Gemini-1.5-Pro**	**Qwen2.5-7B-Instruct**	$\mathbf{62.6}^{+0.2}_{-0.3}$	$\mathbf{31.5}^{+0.2}_{-0.3}$	**46.8**	**17.9**

Table 3: Arithmetic differences from the reported results (percentage points). These are not new measurements or significance estimates.

Comparison	ΔSR	ΔSPL
HGWM–WMNav, HM3D v0.1	+4.5	+0.3
HGWM–WMNav, MP3D	+1.4	+0.7
Voxel–no memory, HM3D v0.2	+3.1	+0.8
HGWM–voxel, HM3D v0.2	+3.5	+1.8

Table 4: Two-backbone sensitivity on HM3D v0.1. Changes from the default are in percentage points (pp).

VLM Backbone	SR (%)	SPL (%)
Gemini-1.5-Pro	**62.6**	**31.5**
Gemini 2.0 Flash	59.2	30.8
Change (pp)	−3.4	−0.7

refinement, or arbitration. A same-prior comparison must retain these components and specify how matching-dependent phases are controlled, as detailed in the supplement.

Success, efficiency, and attribution. Full HGWM improves over voxel memory by 3.5 SR and 1.8 SPL points, changing representation and decision making together. Separately, the author reports gains of 1.5 SR and 1.4 SPL points for scene–goal graph matching over prior prompting. This comparison addresses the explicit compatibility mechanism; its absolute arm scores, split, and detailed run configuration are not separately supplied. The supplement distinguishes this reported gain from the conditions needed for controlled attribution. Neither aggregate differences nor the available query records establish significance or equal inference cost.

4.7 Efficiency and Limitations

The reported implementation requires approximately 2.3 seconds per Gemini-1.5-Pro inference step. The supplement reports graph-update times of 12–25 ms at the tested episode lengths. These are limited runtime measurements, not a complete end-to-end deployment benchmark: hardware, timing variability, and total evaluation cost are not fully reported. Episode-local prior reuse reduces repeated queries; the supplied code does not establish deterministic replay across complete episodes. Overlapping inference with robot motion is a possible deployment strategy, not an evaluated result.

Query and latency accounting. One panoramic response covers six headings; prior generation, relation reasoning, arbitration, and planning can add calls. Thus per-call latency is not full-step latency. The revised runner records client calls and time by role. Provider-internal retries, token usage, simulation time, and motion time still require separate accounting; using the same evaluation episodes does not ensure equal inference cost.

Evidence and robustness limits. Evaluation is confined to Habitat. There are no physical-robot trials, controlled

Table 5: Author-reported HM3D v0.2 results, including corrected voxel-memory and no-refinement SPL. Metrics are percentages; differences are percentage points from full HGWM. The first two baselines change the overall configuration. The non-adaptive control's exact static/uniform mode is not archived.

Configuration	SR (%)	SPL (%)	ΔSR	ΔSPL
Basic VLM / no memory	67.8	29.3	−6.6	−2.6
Voxel memory + direct VLM scoring	70.9	30.1	−3.5	−1.8
HGWM (full)	**74.4**	**31.9**	0.0	0.0
Without goal prior	72.1	30.8	−2.3	−1.1
Without relation refinement	72.1	30.4	−2.3	−1.5
Without LLM arbitration	72.7	31.1	−1.7	−0.8
With non-adaptive fine weights (reported)	73.5	31.3	−0.9	−0.6

localization/depth-noise tests, dynamic-scene tests, or relation-refinement error measurements. Therefore, real-world robustness and correction of perceptual hallucinations remain unverified. Room–object priors may reject legitimate unusual layouts; erroneous detections and pose estimates can persist in memory. The system also depends on foundation-model availability and latency. The point estimates and component ablations do not remove the need for uncertainty estimates or tests of unusual layouts and noisy sensors.

5 Conclusion

HGWM expands category goals into relational priors and combines observed graph memory with scoring on a 0–10 decision scale. Simulation results show SR gains over WMNav and smaller SPL gains. Additional HM3D v0.2 interventions reduce SR by 0.9–2.3 points and SPL by 0.6–1.5 points relative to full HGWM. These findings support the combined design; component interactions, paired uncertainty, query cost, and deployment beyond simulation remain open evaluation questions.

Code Availability. Code, prompt builders, repair tests, and the case gallery are available at `https://github.com/itak04/HGWM` and `https://itak04.github.io/HGWM/`.

Supplementary Material

A Goal Prior Construction

A.1 LLM Prompt Template

We present the high-level strategy template for goal subgraph construction; the source index identifies the executable builder, including its output schema and examples. The template is dynamically instantiated for each target object category, with `{goal}` replaced by the specific target (e.g., "bed", "sofa").

> You are an expert in residential navigation and spatial understanding. Create a comprehensive navigation strategy for efficiently locating a {GOAL} in a typical home environment.
>
> NAVIGATION STRATEGY ANALYSIS FOR {GOAL}:
>
> 1. TARGET ROOM ANALYSIS:
> (1) Primary room: The most likely room type where {goal} is typically found
> (2) Secondary rooms: 1-2 alternative room types where {goal} might also be located
> (3) Adjacent zones: Connected areas that provide visual access or pathways to target rooms
>
> 2. PATHWAY and CONNECTIVITY MAPPING:
> (1) Hallway connections: Common routes to reach target rooms from main areas
> (2) Visual access points: Doorways/openings that help identify target rooms
> (3) Navigation landmarks: Key reference points for wayfinding
>
> 3. TARGET-ORIENTED SPATIAL ANCHORS and NAVIGATION CUES:
> (1) Directional anchors: 3-4 objects that DIRECTLY indicate where {goal} is likely located
> (2) Visual pathways: Objects that form visual lines leading toward typical {goal} placement areas
> (3) Distance indicators: Objects that help estimate proximity to {goal}
> (4) Positioning cues: Furniture/fixtures indicating the specific area where {goal} is usually found
>
> 4. SYSTEMATIC EXPLORATION APPROACH:
> (1) Exploration priorities: Which areas to check first when searching
> (2) Recovery strategies: What to do when initial search paths fail

This structured prompt enables the LLM to generate hierarchical spatial knowledge encompassing room classifications, connectivity patterns, and object-based navigation cues, which are then parsed into the formal goal subgraph representation $G_{goal} = (V_{goal}, E_{goal})$ where nodes represent rooms/objects and edges encode spatial relationships.

B Implementation Details

B.1 Agent Configuration

We configure the agent with a maximum of 40 navigation steps per episode. The agent has a cylindrical body with radius 0.18m and height 0.88m, equipped with an egocentric RGB-D camera at 640×480 resolution and 79° horizontal FOV. The camera is tilted down with a 14° pitch to improve navigable area identification. The success threshold d_{thres} is set to 1.0m. Gemini-1.5-Pro is the primary VLM used for the reported experiments.

B.2 Model Choices

VLM Backbone: Gemini-1.5-Pro serves as our primary vision-language model for both scene understanding (object detection, room classification) and coarse semantic scoring. The reported experiments do not include a matched cost or spatial-reasoning comparison against GPT-4V.

LLM Reasoner: Qwen2.5-7B-Instruct supplies goal priors, relation proposals, and direction recommendations. The default source configuration accesses it through a remote SiliconFlow client; local deployment was not established for the historical measurements. Separate end-to-end reasoner latency is not reported.

C Coarse-to-Fine Reasoning

C.1 Score Domains and Phase Selection

For the six panorama headings, the coarse VLM prompt requests scores in $[0, 10]$. A model-reported visible target receives 10; an unpromising dead end without a visible target receives 0; intermediate opportunities receive 1–9. Missing individual scores default to 5, and failed directional extraction defaults all six headings to 5. A target score is not a ground-truth detection.

The fine overlap is $f^{\alpha} \in [0, 1]$ in the intended score path. Let $\bar{f}$ and f^{max} denote the mean and maximum over headings. The implemented schedule is ordered as follows:

1. If $f^{max} \geq 0.7$, use target verification.
2. Otherwise, if $\bar{f} \geq 0.4$ or $f^{max} \geq 0.5$, use focused search.
3. Otherwise, use frontier exploration.

The main paper lists the three coefficient triples for verification, focused search, and exploration. The source names them spatial, semantic, and LLM weights. Here the first coefficient multiplies the coarse VLM score, despite its name; it does not multiply a separate depth score.

The arbitrator returns a recommended heading $\hat{\alpha}$ and a scalar u. The boost is u for the recommended heading and $\max(0.2, u/2)$ elsewhere. Fusion multiplies the fine score and LLM boost by 10, adds the weighted coarse score, clips to $[0, 10]$, and rounds to two decimal places. A coarse score at least 10 overrides this rule with 10. The coefficients do not

sum to one. No candidate-set or episode-level normalization, estimated confidence bounds, or additional position-score multiplier is applied by this routine. Equal directional scores therefore require no division by a zero range or variance. Final heading selection also depends on the stored curiosity map.

C.2 Symbolic Feature Definitions

All name-based set comparisons below lowercase strings. Define $J(A,B) = |A \cap B| / \max(1, |A \cup B|)$, so two empty sets have overlap zero. These are symbolic matching operations, not cosine similarity of learned embeddings.

Room feature s_r. The prior room set contains names in its `nodes` list with one of the substrings kitchen, bathroom, bedroom, living, dining, or office. The observed set uses `room_nodes`. Known spatial-region labels form a second pair of sets. The score is $0.5J(\text{rooms}) + 0.3J(\text{regions}) + 0.2C$, capped at one, where C is the fraction of room pairs passing the supplied room-connection predicate. For fewer than two observed rooms, the repaired implementation sets $C = 0$. The remaining connectivity predicate is heuristic, not a verified floor-plan relation.

Object feature s_o. Let V be current VLM object names, H observed graph object names, and T the union of the requested category, prior anchors, and named prior object nodes. Let $P \subseteq H$ contain objects in matched rooms; if no rooms match, all observed graph objects are prioritized. Put $n = |V \cap T|$ and

$$v = \min\{1, J(V,T) + \min(0.6, 0.5n) + 0.2\mathbb{I}[T \neq \varnothing,\ T \subseteq V]\}.$$

with $v = 0$ when $n = 0$. The stored-object score is

$$h = \min\{1, J(H,T) + 0.2|P \cap T| / \max(1, |T|)\}.$$

The object feature is $\min\{1, 0.7v + 0.2h + 0.1J(V,H) + 0.1s_r\}$ if $n > 0$, and $\min\{1, 0.4v + 0.4h + 0.2J(V,H) + 0.1s_r\}$ otherwise. The source's `target_detected` flag means any member of T matches, including an anchor; it is not exclusively detection of the navigation target.

Spatial feature s_p. The repaired scorer uses $s_p = 0.4r + 0.4o + 0.2x$. The three terms compare room, object, and containment relation sets respectively. A relation key consists of canonical endpoint names and a relation label, so different instance IDs do not prevent matching. Known symmetric relations (near, adjacent, connected) ignore endpoint order. Each score is the intersection size divided by the larger set size; a missing observed or expected set gives zero. This replaces defaults that could previously contribute positive evidence when typed graph fields were absent.

Semantic and depth features. The semantic feature is the capped weighted sum

$$s_m = \min\{1, 0.4A + 0.3R + 0.3G\},$$

where A measures VLM–graph semantic alignment, R measures hierarchical semantic richness, and G measures goal relevance. Specifically, $A = J(V,H)$ with $A = 0$ for two empty sets. For room, object, and total edge counts n_r, n_o, n_e, richness is

$$R = \tfrac{1}{2}\left[\min\{1, (0.3n_r + 0.1n_o)/10\} + \min\{1, n_e/20\}\right].$$

Goal relevance divides the number of matching observed objects by the number of named prior nodes, capped at one. Exact name matches count as 1 and predefined semantic-group matches as 0.5. No named prior nodes gives $G = 0.5$. The groups are seating/work furniture; bedding; kitchen fixtures; bathroom fixtures; and books/shelves. The exact group members and predicates are listed in the source index. Exceptions in these three subterms return 0.5. These are deterministic heuristics, not a learned semantic model. For foreground, mid-ground, and background proportions p_k, the depth term is $s_d = \frac{1}{3}\sum_k (1 - |p_k - q_k|)$ with $\mathbf{q} = (0.3, 0.4, 0.3)$. When no depth-layer observations are present, the repaired code returns $s_d = 0$ rather than a near-reference default distribution.

C.3 Fine Weights, Bonuses, and Direction Factors

The first satisfied row of Table 6 determines the weight vector. Scene quality is

$$Q = 0.4(0.6v_{frac} + 0.4h_{frac}) + 0.3\bar{c}_{rel} + 0.3\bar{c}_{room},$$

capped at one. Here v_{frac} and h_{frac} are the fractions of voxel-validated and high-precision objects, and the two means summarize relation and room confidence fields. Terms with no supporting objects/relations/rooms contribute zero.

The additive bonus has three independently applicable terms: (i) $\min(0.15, 0.2s_o s_d)$ if $0.3 \le s_r < 0.8$, $s_o > 0.5$, and $s_d > 0.6$; (ii) $\min(0.15, 0.15s_r + 0.1Q)$ if $0.3 \le s_o < 0.8$ and $s_r > 0.5$; and (iii) $\min(0.1, 0.1\bar{c}_{rel})$ if relations exist and their mean confidence exceeds 0.7.

For a stored map cell i at distance $r_i < 200$ map pixels, let δ_i be the wrapped angular difference to the candidate heading, $v_i = \max(0.1, 1 - r_i/200)$, and $n_i = \min(1, N_i/5)$ for observation count N_i. Direction relevance is

$$d^{\alpha} = \frac{\sum_i \max(0, 1 - |\delta_i|/\pi) v_i n_i}{\max(0.1, \sum_i v_i n_i)}.$$

It defaults to 1 without stored semantic memory or an available agent pose. The depth-direction factor is $h^{\alpha} = \min(1.2, p_F u_F + p_B u_B + 1 - p_F - p_B)$: $(u_F, u_B) = (1.2, 0.8)$ for 30/330 degrees, $(0.9, 1.1)$ for 90/270 degrees, and $(1, 1)$ otherwise. It defaults to 1 without depth-layer objects. The fine routine receives its direction argument from the caller; the primary scoring caller converts numeric degree strings directly to degree values.

C.4 Memory, Prompts, and Execution

The default configuration specifies a navigation map scale of 100 pixels per meter, high-level horizon 40, RGB-D resolution

Table 6: Fine-feature weights from the supplied scoring routine, in first-match order. These are implementation settings, not fitted performance results.

Condition	Room	Object	Spatial	Semantic	Depth
Any prior-associated object detected	0.25	0.50	0.20	0.03	0.02
Scene quality $Q > 0.8$	0.35	0.30	0.25	0.05	0.05
Room score $s_r > 0.7$	0.45	0.25	0.20	0.05	0.05
Object score $s_o > 0.7$	0.30	0.40	0.20	0.05	0.05
Otherwise	0.35	0.30	0.20	0.10	0.05

640×480, horizontal field of view 79°, camera pitch -0.25 radians, body radius 0.18 m, body height 0.88 m, and success distance 1.0 m. These settings do not determine the number of successful evaluated episodes by themselves.

Goal priors are generated at runtime, cached by goal, and cleared on episode reset. No cross-episode canonical-prior archive or persistent image–prompt response cache is supplied. Provider-specific sampling settings are in `src/api.py`. The original claims of universally fixed temperature-zero priors and deterministic identical-query replay are therefore not retained.

The goal prompt printed earlier is a readable strategy template. Executable prompt builders additionally define output schemas, examples, and error handling. The accompanying code contains the exact strings used for prior construction, panorama scoring, relation proposals, arbitration, and local planning. The repository's source index maps these builders to source locations so that the paper does not substitute an abbreviated prompt for the executable one.

C.5 Implementation Repairs and Offline Validation

The code now normalizes legacy and hierarchical scene fields at the scoring boundary, includes both retrieved memory and current evidence, and exposes synchronized flat and typed edge collections. Object room aliases and goal-prompt aliases are normalized. Relation edits cannot leave stale edges in another view. Endpoint names, rather than incidental instance IDs, determine relation matching. Missing observations, relations, and depth data no longer receive positive match defaults.

Depth-ray positions are queued until the panoramic VLM response is available. Object descriptions supplied as dictionaries or lists are then associated with the closest view-center ray, and stored in persistent memory. These associations are coarse view-level estimates; they are not precise visual object localization. Planning prompts read room dictionaries correctly and express heading degrees once.

The 28 offline tests exercise graph conversion, relation updates and matching, memory ingestion, absent evidence, fallbacks, planning, ablations, call timing, run records, and statistical summaries. Small fixtures execute source methods without a simulator or model API. These tests verify software behavior. Historical cases predate the repairs. The author reports unchanged aggregate performance after repair, with two SPL corrections; source-linked episode records are still needed for verification.

Table 7: Runtime values retained from the original report; hardware and cache conditions were not established for this revision.

Episode length	10	20	40
Memory (MB)	1.2	2.8	5.1
Graph update (ms)	12	18	25
VLM inference (ms)	2300	2300	2300

C.6 Reevaluation Controls and Recorded Outputs

Uniform fine weights assign 0.2 to each feature. Static weights normalize a configured vector; the supplied default is

$$(0.30, 0.40, 0.20, 0.05, 0.05),$$

ordered by room, object, spatial, semantic, and depth features. These are separate modes. The historical 73.5/31.3 result has no archived exact mode.

The original no-arbitration flag preserves unnormalized removal for comparison with the reported result. A separate matched-scale mode rescales the remaining coefficients by $(a+b+e)/(a+b)$. The prior-prompt-only control bypasses graph scoring and observed-graph planning context. The graph-memory/direct-scoring control preserves graph context but uses the raw visual direction scores. No outcome is supplied for these newly prepared controls.

Runs save manifests with source hashes, configuration, and seed; selected episode identities; client-call counts and times by role; graph diagnostics; and JSON episode outcomes. Client meters exclude provider-internal retries and token counts. The analysis script matches dataset, split, scene, episode, and seed, rejects omitted episodes, and resamples paired scene clusters. With fewer than two scenes it returns no interval. No confidence interval was inferred from historical aggregates.

C.7 Reported Runtime Measurements

These measurements do not establish end-to-end latency, asymptotic scaling, or total evaluation cost.

Table 8: Reported configurations on HM3D v0.2 [18]. The final row changes both memory and scoring; it does not isolate their effects or measure ambiguity robustness.

Configuration	**SR(%)↑**	**SPL(%)↑**
Basic VLM / no memory	67.8	29.3
VLM / voxel memory	70.9	30.1
HGWM / graph matching	**74.4**	**31.9**

Table 9: Additional HM3D v0.2 component results.

Configuration	SR (%)	SPL (%)
Full HGWM	74.4	31.9
Without goal prior	72.1	30.8
Without relation refinement	72.1	30.4
Without LLM arbitration	72.7	31.1
With non-adaptive fine weights	73.5	31.3

D Additional Analyses

D.1 Roles of Coarse and Fine Stages

The stages are intended to rank directions and check compatibility with the prior. The current component comparison does not separately measure coarse-stage recall, fine-stage precision, pruning error rates, or loop frequency. It therefore cannot explain the results through a proven division of labor. In particular, prior consistency is not a ground-truth test: correct but unusual observations can be rejected, and plausible errors can remain.

D.2 Full Component Configuration

Table 8 restates the three reported configurations. Basic VLM uses no persistent memory; the voxel configuration retains direct VLM scoring; HGWM changes both memory and decision scoring.

D.3 Additional Component Results

The additional HM3D v0.2 results are expressed relative to full HGWM at 74.4% SR and 31.9% SPL. Table 9 records the resulting absolute values. The non-adaptive result is retained as one historical control; its exact static/uniform source mode has not been established. The existing 70.9/30.1 direct-scoring baseline remains labeled voxel memory.

The reported decreases are 2.3/1.1 points for no prior, 2.3/1.5 for no relation refinement, 1.7/0.8 for no arbitration, and 0.9/0.6 for the non-adaptive weight control. These values support comparison with the full model at the level of aggregate point estimates. They do not provide paired confidence intervals, interactions between removals, or a per-query cost comparison.

Table 10: Reported comparisons relevant to matching attribution. Gains and drops are percentage-point differences, not absolute SR/SPL. The scene–goal matching gain is author-reported; its detailed run configuration is not supplied.

Quantity	Evidence
Full vs. voxel-memory SR/SPL gains	3.5/1.8 pp
Removal of refinement SR/SPL drops	2.3/1.5 pp
Scene–goal matching vs. prompting: SR/SPL gains	+1.5/+1.4 pp

D.4 Prior Information and Matching: Attribution Protocol

The goal prior has two roles: text conditioning for the coarse VLM and a relational reference for symbolic scoring. Both can influence behavior; a prompt can reason about relations without executing HGWM's matching routine. The proposed novelty is an explicit prior-to-observation computation connected to direction ranking, not privileged prior information or a claim that prompting cannot use structure.

Matched online comparison. The author reports gains of 1.5 SR and 1.4 SPL percentage points for explicit scene–goal graph matching over prior prompting. These are differences, not absolute scores for either arm. The comparison split, individual arm scores, and run configuration have not been separately supplied. The conditions below define the intended controlled comparison; their fulfillment by the reported run cannot be established from the aggregate differences alone.

Both arms must share the episode set, model versions, sampling settings, and cached goal prior for each paired episode. Keep memory construction, relation refinement, access to prior and graph context, map updates, stopping rules, and the local action interface unchanged.

Use one common phase rule defined independently of fine scores. Keep the arbitration prompt template identical and omit fine scores from both arms' arbitration inputs. One arm uses fine matching in fusion; the other uses the equal-coefficient coarse substitution above. Both retain clipping and the coarse-score override. These are new controlled configurations, not replacements for the reported full-model row. Observations and model responses may diverge after actions differ; they are not held equal during navigation. Record calls, available token counts, and elapsed time to assess cost.

Relation to the supplied switches. The existing prior-prompt-only switch also removes persistent graph context, refinement, and arbitration. The graph-memory/direct switch preserves graph context but removes refinement, fine scoring, and arbitration. Neither is the matched comparison just defined. The no-prior and no-refinement rows answer narrower intervention questions; they cannot substitute for a direct test of explicit matching. The supplied runner does not implement the new matched-online protocol as a dedicated mode.

D.5 Three-Run Range on HM3D v0.1

The author reports three completed runs for the main HGWM configuration using Gemini-1.5-Pro on HM3D v0.1. Relative to the tabulated SR of 62.6 and SPL of 31.5, the maximum and minimum deviations are $+0.2$ and -0.3 percentage points for both metrics. Thus the observed SR range is 62.3–62.8 and the SPL range is 31.2–31.7. These are marginal ranges: the extrema need not occur in the same run. The three individual result pairs and whether the tabulated point is the mean were not specified, so no standard deviation, standard error, or confidence interval is derived. Seeds and episode-set identity across runs are not supplied. The range does not extend to MP3D, HM3D v0.2, the Flash backbone, or the baseline methods. In particular, it does not establish a paired significant improvement over WMNav.

D.6 Further Evaluation

An optional diagnostic could replay recorded states with the same prior, graph, coarse scores, phase, and LLM output, replacing $10bf^{\alpha}$ with bc^{α} as defined in the main paper. Comparing rankings before and after clipping and map updates would characterize the direct numerical matching channel. Such replay retains any matching information already present in the phase and LLM output; it cannot estimate closed-loop SR or SPL. No replay result is reported here.

Useful remaining comparisons include graph versus voxel memory under the same direct VLM decision rule, coarse-only and fine-only scoring, and joint component removals. Record identical episode IDs and source versions, and account for prompts, seeds, and model-query budgets. Retained/changed relation annotations could measure whether refinement introduces or removes errors; loop frequency, revisitation, and failure categories would complement SR/SPL. These analyses are not supplied by the four aggregate ablation rows.

Robustness protocol still needed. Test controlled depth and localization perturbations, unusual room–object arrangements, and dynamic scenes separately from backbone changes. Physical-robot trials should report platform and sensors, trial counts, safety controls, success criteria, collision/failure rates, and end-to-end latency. No such evaluations are reported in this study.

E Limitations and Future Work

Current Limitations.

- *Simulation-only Evidence:* No physical-robot results or controlled noise robustness tests are available. Backbone sensitivity is evaluated for only two VLM settings on one split.
- *Component Attribution:* The memory baselines change representation and scoring together. Additional component results are available, but their interactions and statistical uncertainty remain unestablished.
- *Static Environments:* Our current framework assumes static scenes. Dynamic objects (e.g., people, pets) or movable furniture could violate the structural priors encoded in G_{goal}.
- *Foundation Model Dependency:* Performance is constrained by the capabilities of the underlying VLMs/LLMs. Detection errors can propagate to graph construction; heuristic pruning can also remove correct atypical observations.
- *Computational Cost:* Despite caching optimizations, real-time deployment remains challenging due to VLM inference latency (2.3s per call).

Future studies could evaluate temporal graph updates and search for multiple targets. Active perception and physical robot trials are also untested directions.

F Source and Evaluation Provenance

The public repository at `https://github.com/itak04/HGWM` contains code, configuration, prompt builders, ablation controls, and analysis scripts. The four additional HM3D v0.2 scores follow the specified decreases from 74.4/31.9; no outcome is forecast. Three-run minimum/maximum deviations are reported for the main HM3D v0.1 setting. Exact experiment-version provenance, episode lists, individual run pairs, hardware, and query totals remain unavailable. The code notes document the repairs and their offline tests. The author reports unchanged post-repair performance and corrects two SPL entries; no source-linked episode records accompany that report. The selected cases are historical traces, not post-repair evaluations. Datasets and models retain their providers' access terms; users supply their own credentials.

LLM roles and writing assistance. Model roles and executable prompts are documented in the code. AI-assisted tools helped edit and format the manuscript; authors supplied the measurements and remain responsible for the scientific claims.

Code availability. The code directory contains a README, implementation notes, source index, and configuration files. The project page links the manuscript, source, and case gallery.

G Archived Navigation Cases

The supplementary archive covers all 18 supplied case folders. It indexes 338 step directories and retains 337 available text logs. Table 11 lists the coverage, including one missing text record. Open the case gallery linked from `https://itak04.github.io/HGWM/` to browse each case and its prompts, responses, selected views, curiosity maps, and navigation maps.

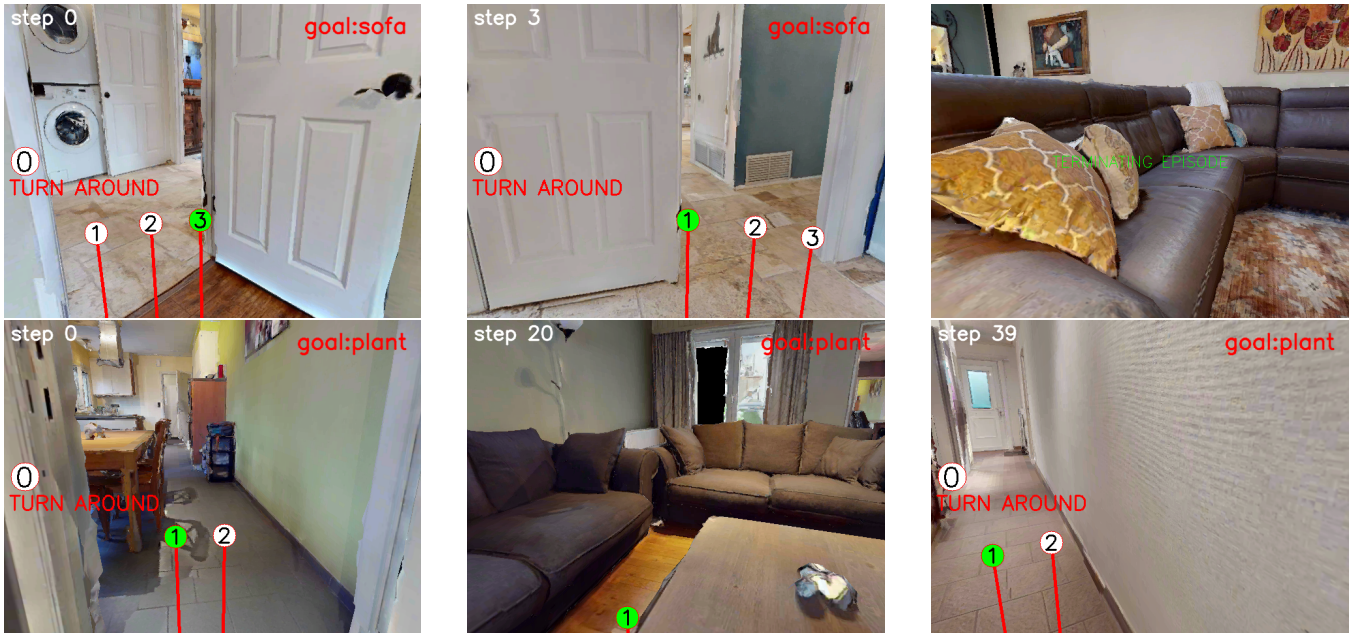


Figure 5: Unedited archived views. Top: case 115_853 at steps 0, 3, and 5. Bottom: case 220_890 at steps 0, 20, and 39. Text and action annotations are part of the original images. The examples illustrate a short and a longer recorded trace; they are not paired method comparisons or repaired-code evaluations.

Selection and fidelity. The original collection is about 913 MB. To fit the supplementary limit, the first, middle, and last available step of each folder are selected by one deterministic rule, without consulting outcomes. The selected PNG files are copied byte-for-byte. All available step text is preserved, with a manifest recording source checksums and inclusion decisions. Additional frames and GIFs are omitted; pickle files are not deserialized or required by the gallery. The original source directory is unchanged.

What these records establish. The logs expose what the model was prompted with and what it returned. For example, case 115_853 begins with an open-door subtask and ends with a logged sofa locator response; the final image visibly contains a sofa. Such observations help inspect a decision sequence, but a model goal flag or termination overlay is not a verified simulator success value. Some logged prior summaries contain zero node/edge counts; these summaries are retained as written and are not assumed to fully describe a hidden runtime graph.

The text records do not establish the exact benchmark split, model version, or per-episode simulator SR/SPL. No aggregate navigation score or error-frequency estimate is computed from this author-selected collection. Error-marked folders are retained rather than filtered out. These historical records do not validate the new graph-contract code or its control variants; their purpose is qualitative traceability.

Table 11: Coverage of the supplied case archive. Counts describe archived steps and text records, not successful episodes.

Case	Logged target	Steps	Logs
115_853	sofa	6	6
160_853	tv screen	7	7
199_890	plant	29	29
220_890	plant	40	40
249_890	tv screen	40	40
265_890	toilet	19	19
289_890	chair	40	40
297_843	toilet	8	8
2_877	bed	4	4
310_843	sofa	2	2
32_877	chair	6	6
352_843	toilet	15	15
382_843	bed	24	24
399_824	bed	14	14
450_824	bed	40	40
546_880	chair	4	4
594_829	chair	10	9
63_877	sofa	30	30

References

[1] Peter Anderson, Angel X. Chang, Devendra Singh Chaplot, Alexey Dosovitskiy, Saurabh Gupta, Vladlen Koltun, Jana Kosecka, Jitendra Malik, Roozbeh Mottaghi, Manolis Savva, and Amir R. Zamir. On evaluation of embodied navigation agents. *arXiv preprint arXiv:1807.06757*, 2018. doi: 10.48550/arXiv.1807.06757.

[2] Amir Bar, Gaoyue Zhou, Danny Tran, Trevor Darrell, and Yann LeCun. Navigation world models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15791–15801. IEEE, 2025. doi: 10.1109/CVPR52734.2025.01472. URL https://openaccess.thecvf.com/content/CVPR2025/html/Bar_Navigation_World_Models_CVPR_2025_paper.html.

[3] Dhruv Batra, Aaron Gokaslan, Aniruddha Kembhavi, Oleksandr Maksymets, Roozbeh Mottaghi, Manolis Savva, Alexander Toshev, and Erik Wijmans. Objectnav revisited: On evaluation of embodied agents navigating to objects. *arXiv preprint arXiv:2006.13171*, 2020. doi: 10.48550/arXiv.2006.13171.

[4] Wenzhe Cai, Siyuan Huang, Guangran Cheng, Yuxing Long, Peng Gao, Changyin Sun, and Hao Dong. Bridging zero-shot object navigation and foundation models through pixel-guided navigation skill. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 5228–5234. IEEE, 2024. doi: 10.1109/ICRA57147.2024.10610499.

[5] Angel X. Chang, Angela Dai, Thomas Funkhouser, Maciej Halber, Matthias Niessner, Manolis Savva, Shuran Song, Andy Zeng, and Yinda Zhang. Matterport3D: Learning from RGB-D data in indoor environments. In *Proceedings of the International Conference on 3D Vision*, pages 667–676. IEEE, 2017. doi: 10.1109/3DV.2017.00081.

[6] Junting Chen, Guohao Li, Suryansh Kumar, Bernard Ghanem, and Fisher Yu. How to not train your dragon: Training-free embodied object goal navigation with semantic frontiers. In *Robotics: Science and Systems XIX*. Robotics: Science and Systems Foundation, 2023. doi: 10.15607/RSS.2023.XIX.075. URL https://roboticsproceedings.org/rss19/p075.html.

[7] Andy Clark. Whatever next? predictive brains, situated agents, and the future of cognitive science. *Behavioral and brain sciences*, 36(3):181–204, 2013. doi: 10.1017/S0140525X12000477.

[8] Karl Friston. The free-energy principle: a unified brain theory? *Nature reviews neuroscience*, 11(2):127–138, 2010. doi: 10.1038/nrn2787.

[9] Samir Yitzhak Gadre, Mitchell Wortsman, Gabriel Ilharco, Ludwig Schmidt, and Shuran Song. CoWs on pasture: Baselines and benchmarks for language-driven zero-shot object navigation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23171–23181. IEEE, 2023. doi: 10.1109/CVPR52729.2023.02219. URL https://openaccess.thecvf.com/content/CVPR2023/html/Gadre_CoWs_on_Pasture_Baselines_and_Benchmarks_for_Language-Driven_Zero-Shot_Object_CVPR_2023_paper.html.

[10] Qiao Gu, Alihusein Kuwajerwala, Sacha Morin, Krishna Murthy Jatavallabhula, Bipasha Sen, Aditya Agarwal, Corban Rivera, William Paul, Kirsty Ellis, Rama Chellappa, Chuang Gan, Celso Miguel de Melo, Joshua B. Tenenbaum, Antonio Torralba, Florian Shkurti, and Liam Paull. ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 5021–5028. IEEE, 2024. doi: 10.1109/ICRA57147.2024.10610243.

[11] David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018. doi: 10.48550/arXiv.1803.10122.

[12] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640:647–653, 2025. doi: 10.1038/s41586-025-08744-2.

[13] I-Tak Ieong and Hao Tang. Multimodal perception for goal-oriented navigation: A survey. *arXiv preprint arXiv:2504.15643*, 2025. doi: 10.48550/arXiv.2504.15643.

[14] Jing Yu Koh, Honglak Lee, Yinfei Yang, Jason Baldridge, and Peter Anderson. Pathdreamer: A world model for indoor navigation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 14718–14728. IEEE, 2021. doi: 10.1109/ICCV48922.2021.01447. URL https://openaccess.thecvf.com/content/ICCV2021/html/Koh_Pathdreamer_A_World_Model_for_Indoor_Navigation_ICCV_2021_paper.html.

[15] Yuxuan Kuang, Hai Lin, and Meng Jiang. OpenFMNav: Towards open-set zero-shot object navigation via vision-language foundation models. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 338–351, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-naacl.24. URL https://aclanthology.org/2024.findings-naacl.24/.

[16] Arjun Majumdar, Gunjan Aggarwal, Bhavika Devnani, Judy Hoffman, and Dhruv Batra. ZSON: Zero-shot object-goal navigation using multimodal goal embeddings. In *Advances in Neural Information Processing Systems*, volume 35, pages 32340–32352, 2022. doi: 10.52202/068431-2343. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/d0b8f0c8f79d3a621af945cafb669f4b-Abstract-Conference.html.

[17] Dujun Nie, Xianda Guo, Yiqun Duan, Ruijun Zhang, and Long Chen. WMNav: Integrating vision-language models into world models for object goal navigation. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 2392–2399. IEEE, 2025. doi: 10.1109/IROS60139.2025.11246684.

[18] Santhosh K. Ramakrishnan, Aaron Gokaslan, Erik Wijmans, Oleksandr Maksymets, Alexander Clegg, John Turner, Eric Undersander, Wojciech Galuba, Andrew Westbury, Angel X. Chang, Manolis Savva, Yili Zhao, and Dhruv Batra. Habitat-matterport 3D dataset (HM3D): 1000 large-scale 3D environments for embodied AI. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1, 2021. URL https://datasets-benchmarks-proceedings.neurips.cc/paper_files/paper/2021/hash/34173cb38f07f89ddbebc2ac9128303f-Abstract-round2.html.

[19] Ram Ramrakhya, Eric Undersander, Dhruv Batra, and Abhishek Das. Habitat-Web: Learning embodied object-search strategies from human demonstrations at scale. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5163–5173. IEEE, 2022. doi: 10.1109/CVPR52688.2022.00511. URL https://openaccess.thecvf.com/content/CVPR2022/html/Ramrakhya_Habitat-Web_Learning_Embodied_Object-Search_Strategies_From_Human_Demonstrations_at_Scale_CVPR_2022_paper.html.

[20] Jingwen Sun, Jing Wu, Ze Ji, and Yu-Kun Lai. A survey of object goal navigation. *IEEE Transactions on Automation Science and Engineering*, 22:2292–2308, 2025. doi: 10.1109/TASE.2024.3378010.

[21] Xinyu Sun, Lizhao Liu, Hongyan Zhi, Ronghe Qiu, and Junwei Liang. Prioritized semantic learning for zero-shot instance navigation. In *Computer Vision – ECCV 2024*, volume 15070 of *Lecture Notes in Computer Science*, pages 161–178. Springer Nature Switzerland, 2025. doi: 10.1007/978-3-031-73254-6_10. URL https://doi.org/10.1007/978-3-031-73254-6_10.

[22] Pengying Wu, Yao Mu, Bingxian Wu, Yi Hou, Ji Ma, Shanghang Zhang, and Chang Liu. VoroNav: Voronoi-based zero-shot object navigation with large language model. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 53757–53775. PMLR, 2024. URL https://proceedings.mlr.press/v235/wu24u.html.

[23] Karmesh Yadav, Ram Ramrakhya, Arjun Majumdar, Vincent-Pierre Berges, Sachit Kuhar, Dhruv Batra, Alexei Baevski, and Oleksandr Maksymets. Offline visual representation learning for embodied navigation. *arXiv preprint arXiv:2204.13226*, 2022. doi: 10.48550/arXiv.2204.13226.

[24] Karmesh Yadav, Arjun Majumdar, Ram Ramrakhya, Naoki Yokoyama, Alexei Baevski, Zsolt Kira, Oleksandr Maksymets, and Dhruv Batra. OVRL-V2: A simple state-of-art baseline for ImageNav and ObjectNav. *arXiv preprint arXiv:2303.07798*, 2023. doi: 10.48550/arXiv.2303.07798.

[25] Hang Yin, Xiuwei Xu, Zhenyu Wu, Jie Zhou, and Jiwen Lu. SG-Nav: Online 3D scene graph prompting for LLM-based zero-shot object navigation. In *Advances in Neural Information Processing Systems*, volume 37, pages 5285–5307, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/098491b37deebbe6c007e69815729e09-Abstract-Conference.html.

[26] Hang Yin, Xiuwei Xu, Linqing Zhao, Ziwei Wang, Jie Zhou, and Jiwen Lu. UniGoal: Towards universal zero-shot goal-oriented navigation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19057–19066. IEEE, 2025. doi: 10.1109/CVPR52734.2025.01775. URL https://openaccess.thecvf.com/content/CVPR2025/html/Yin_UniGoal_Towards_Universal_Zero-shot_Goal-oriented_Navigation_CVPR_2025_paper.html.

[27] Zecheng Yin, Hao Zhao, and Zhen Li. Hypernav: Hybrid perception for object-oriented navigation in unknown

environment. *arXiv preprint arXiv:2510.22917*, 2025. doi: 10.48550/arXiv.2510.22917.

[28] Naoki Yokoyama, Sehoon Ha, Dhruv Batra, Jiuguang Wang, and Bernadette Bucher. VLFM: Vision-language frontier maps for zero-shot semantic navigation. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 42–48. IEEE, 2024. doi: 10.1109/ICRA57147.2024.10610712.

[29] Bangguo Yu, Hamidreza Kasaei, and Ming Cao. L3MVN: Leveraging large language models for visual target navigation. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 3554–3560. IEEE, IEEE, 2023. doi: 10.1109/IROS55552.2023.10342512.

[30] Jiazhao Zhang, Liu Dai, Fanpeng Meng, Qingnan Fan, Xuelin Chen, Kai Xu, and He Wang. 3D-aware object goal navigation via simultaneous exploration and identification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6672–6682. IEEE, 2023. doi: 10.1109/CVPR52729.2023.00645. URL https://openaccess.thecvf.com/content/CVPR2023/html/Zhang_3D-Aware_Object_Goal_Navigation_via_Simultaneous_Exploration_and_Identification_CVPR_2023_paper.html.

[31] Mingjie Zhang, Yuheng Du, Chengkai Wu, Jinni Zhou, Zhenchao Qi, Jun Ma, and Boyu Zhou. Apexnav: An adaptive exploration strategy for zero-shot object navigation with target-centric semantic fusion. *IEEE Robotics and Automation Letters*, 10(11):11530–11537, 2025. doi: 10.1109/LRA.2025.3606388. URL https://doi.org/10.1109/LRA.2025.3606388.

[32] Sixian Zhang, Xinhang Song, Yubing Bai, Weijie Li, Yakui Chu, and Shuqiang Jiang. Hierarchical object-to-zone graph for object navigation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 15110–15120. IEEE, 2021. doi: 10.1109/ICCV48922.2021.01485. URL https://openaccess.thecvf.com/content/ICCV2021/html/Zhang_Hierarchical_Object-to-Zone_Graph_for_Object_Navigation_ICCV_2021_paper.html.

[33] Sixian Zhang, Xinyao Yu, Xinhang Song, Xiaohan Wang, and Shuqiang Jiang. Imagine before go: Self-supervised generative map for object goal navigation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16414–16425. IEEE, 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Zhang_Imagine_Before_Go_Self-Supervised_Generative_Map_for_Object_Goal_Navigation_CVPR_2024_paper.html.

[34] Linqing Zhong, Chen Gao, Zihan Ding, Yue Liao, Huimin Ma, Shifeng Zhang, Xu Zhou, and Si Liu. TopV-Nav: Unlocking the top-view spatial reasoning potential of MLLM for zero-shot object navigation. *arXiv preprint arXiv:2411.16425*, 2024. doi: 10.48550/arXiv.2411.16425.

[35] Kaiwen Zhou, Kaizhi Zheng, Connor Pryor, Yilin Shen, Hongxia Jin, Lise Getoor, and Xin Eric Wang. ESC: Exploration with soft commonsense constraints for zero-shot object navigation. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 42829–42842. PMLR, PMLR, 2023. URL https://proceedings.mlr.press/v202/zhou23r.html.

[36] Zibo Zhou, Yue Hu, Lingkai Zhang, Zonglin Li, and Siheng Chen. Beliefmapnav: 3d voxel-based belief map for zero-shot object navigation. *arXiv preprint arXiv:2506.06487*, 2025. doi: 10.48550/arXiv.2506.06487.

[37] Fengda Zhu, Xiwen Liang, Yi Zhu, Qizhi Yu, Xiaojun Chang, and Xiaodan Liang. SOON: Scenario oriented object navigation with graph-based exploration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12684–12694. IEEE, 2021. doi: 10.1109/CVPR46437.2021.01250. URL https://openaccess.thecvf.com/content/CVPR2021/html/Zhu_SOON_Scenario_Oriented_Object_Navigation_With_Graph-Based_Exploration_CVPR_2021_paper.html.